

QAOA custom mixer problem

<https://github.com/qiskit-community/qiskit-aqua/issues/1064>

ROW 44

QAOA custom mixer problem #1064



Closed



woodsp-ibm opened on Jun 19, 2020

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This issue was reported on the aqua channel in the Qiskit slack workspace. This is the original report

QAOA XY mixer - I want to create an XY mixer and a simple cost function in QAOA. So that I can confirm, As Wang showed in April 2019, that all parameters will create a state conforming to the XY mixer constraint. In this way the cost function does not need to reflect the constraints because the XY mixer can do this instead.

The programme below Creates either the default X mixer if UseTraditionalMixer == True, or my attempt at the XY mixer otherwise. It runs successfully with the former, but give the Error messages;

TypeError: unsupported operand type(s) for *: 'complex' and 'ParameterExpression'

The code attached below is based on the sample provided there, and shows the reported problem. In initial investigation it seems that the message is a result of the complex coeff in the Paulis of the mixer being multiplied, via primitive_op mul function, with a ParameterExpression, where latter does not support complex.

```
from qiskit import BasicAer
from qiskit.aqua import QuantumInstance
from qiskit.aqua.algorithms import QAOA
from qiskit.aqua.operators import I, X, Y
from qiskit.aqua.components.optimizers import SLSQP
from qiskit.optimization import QuadraticProgram
from qiskit.optimization.algorithms import MinimumEigenOptimizer

qubo = QuadraticProgram()
linearQ19 = {'a': 0, 'b': 0, 'c': 0}
quadraticQ19 = {'a', 'b': -1, ('a', 'c'): -4, ('b', 'c'): -1}
for val in (linearQ19):
    qubo.binary_var(val) # chr(n+1+96))
qubo.minimize(constant=0, linear=linearQ19, quadratic=quadraticQ19)
theMixer = []
useTraditionalMixer = False
if useTraditionalMixer:
    theMixer.append(X*I*I)
    theMixer.append(I*X*I)
    theMixer.append(I*I*X)
else:
    OperatorXY = X.compose(Y)
    theMixer.append(OperatorXY*I*I)
    theMixer.append(OperatorXY*OperatorXY*I)
mixerUsed = sum(theMixer)

qaoa = QAOA(optimizer=SLSQP(), quantum_instance=BasicAer.get_backend('statevector_simulator'),
            mixer=mixerUsed)
meo = MinimumEigenOptimizer(qaoa, penalty=None)
meo_result = meo.solve(qubo)
print(meo_result)
```



Assignees

No one assigned

Labels

type: bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



woodsp-ibm added type: bug on Jun 19, 2020



woodsp-ibm added this to To do in Operators on Jun 19, 2020



Cryoris on Sep 11, 2020

Contributor

...

The operator attempted to use as mixer Hamiltonian is not actually a Hamiltonian as it is not hermitian. The first summand if useTraditionalMixer=False is

$$(X \otimes Y) \wedge I \wedge I = -iZ \wedge I \wedge I$$


which is not hermitian. Or from another point-of-view, if we try to use this operator in a time evolution we obtain $e^{iH} = e^{iZ}$ which is not unitary.

If we supported imaginary time-evolution we could lift this restriction, but this feature is not yet available. We could turn this into a "feature request" issue 🙄

Zoter